



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Objective: Predict Falcon 9 first-stage landing success using exploratory data analysis and machine learning.

Github Link: <https://github.com/abhaysingh7/IBM-Capstone-Project>

Abhay Singh

July12, 2026



Outline

- **Executive Summary** – Project objectives, methodology, key findings, and business impact.
- **Introduction** – Project background, problem statement, and research objectives.
- **Methodology** – Data collection, data wrangling, exploratory data analysis (EDA), interactive visual analytics, and machine learning approach.
- **Results** – Key findings from EDA, SQL analysis, Folium maps, Plotly Dash dashboard, and predictive model performance.
- **Conclusion** – Summary of insights, model evaluation, and project outcomes.
- **Future Work** – Potential improvements, additional features, and advanced machine learning models.

Executive Summary

Project Objective

- Develop a machine learning model to predict whether the Falcon 9 first-stage booster will land successfully, helping estimate launch costs and support decision-making for commercial space missions.

Data Collection

- Collected Falcon 9 launch data from the SpaceX REST API.
- Supplemented the dataset by scraping Falcon 9 launch records from Wikipedia using BeautifulSoup.
- Performed data cleaning, feature engineering, and preprocessing to prepare the dataset for analysis.

Methodology

- Conducted Exploratory Data Analysis (EDA) using Python visualizations and SQL queries.
- Built interactive visualizations using Folium maps and a Plotly Dash dashboard to analyze launch sites, payloads, orbit types, and mission outcomes.
- Trained and evaluated multiple classification models, including Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN), using GridSearchCV for hyperparameter tuning.

Key Findings

- Launch success rates improved significantly over time, reflecting increasing operational experience.
- Launch site, orbit type, payload mass, and flight number were identified as important factors influencing successful landings.
- KSC LC-39A demonstrated the highest launch success rate among the launch sites.
- The Decision Tree classifier achieved the best predictive performance among the evaluated models.

Conclusion

- The project demonstrates that machine learning can effectively predict Falcon 9 first-stage landing success using historical launch data. These insights can help estimate launch costs, improve mission planning, and support data-driven decision-making for future space missions.

Introduction

Project Background and Context

- SpaceX has significantly reduced the cost of space launches by reusing the Falcon 9 first-stage booster. A Falcon 9 launch costs approximately **\$62 million**, compared to over **\$165 million** for many competing launch providers. Since the successful recovery of the first-stage booster is the key factor enabling reusability, predicting whether a booster will land successfully is valuable for estimating mission costs and evaluating launch performance. This project applies data science and machine learning techniques to predict the landing success of Falcon 9 first-stage boosters using historical launch data.

Project Objective

- Develop an end-to-end data science pipeline that collects, cleans, analyzes, and models Falcon 9 launch data to accurately predict whether the first-stage booster will land successfully.

Research Questions

- Which factors have the greatest impact on Falcon 9 first-stage landing success?
- How do features such as **launch site, payload mass, orbit type, flight number, and booster version** influence landing outcomes?
- Which machine learning classification model provides the most accurate prediction of landing success?
- How can the insights from historical launch data support cost estimation and decision-making for future commercial space missions?

Section 1

Methodology

Methodology

Data Collection

- Collected Falcon 9 launch data using the **SpaceX REST API** and **Wikipedia web scraping**.

Data Wrangling

- Cleaned and preprocessed the dataset.
- Applied **One-Hot Encoding** to categorical features.

Exploratory Data Analysis (EDA)

- Analyzed relationships between **launch success**, **payload mass**, **orbit type**, **flight number**, and **launch site** using visualizations and SQL.

Interactive Visual Analytics

- Built **Folium maps** to visualize launch sites.
- Developed a **Plotly Dash dashboard** for interactive exploration of launch data.

Predictive Analysis

- Trained and tuned **Logistic Regression**, **SVM**, **Decision Tree**, and **KNN** models using **GridSearchCV**.
- Evaluated model performance and selected the best classifier for predicting Falcon 9 first-stage landing success.

Data Collection

- The data was collected using various methods
 - Data collection was done using get request to the SpaceX API.
 - Next, we decoded the response content as a Json using `.json()` function call and turn it into a pandas dataframe using `.json_normalize()`.
 - We then cleaned the data, checked for missing values and fill in missing values where necessary.
 - In addition, we performed web scraping from Wikipedia for Falcon 9 launch records with BeautifulSoup.
 - The objective was to extract the launch records as HTML table, parse the table and convert it to a pandas dataframe for future analysis.

Data Collection – SpaceX API

- We used the get request to the SpaceX API to collect data, clean the requested data and did some basic data wrangling and formatting.
- The link to the notebook:

<https://github.com/abhaysingh7/IBM-Capstone-Project/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>

1. Get request for rocket launch data using API

```
In [6]: spacex_url="https://api.spacexdata.com/v4/launches/past"
```

```
In [7]: response = requests.get(spacex_url)
```

2. Use json_normalize method to convert json result to dataframe

```
In [12]: # Use json_normalize method to convert the json result into a dataframe  
# decode response content as json  
static_json_df = res.json()
```

```
In [13]: # apply json_normalize  
data = pd.json_normalize(static_json_df)
```

3. We then performed data cleaning and filling in the missing values

```
In [30]: rows = data_falcon9['PayloadMass'].values.tolist()[0]  
  
df_rows = pd.DataFrame(rows)  
df_rows = df_rows.replace(np.nan, PayloadMass)  
  
data_falcon9['PayloadMass'][0] = df_rows.values  
data_falcon9
```


Data Collection - Scraping

- Collected Falcon 9 launch data from the **SpaceX REST API**.
- Scraped additional historical launch records from **Wikipedia** using **BeautifulSoup**.
- Parsed HTML tables and converted the extracted data into a **Pandas DataFrame** for further analysis.
- Combined both data sources to create a comprehensive dataset for data wrangling and predictive modeling.
- **Notebook:**
GitHub:
<https://github.com/abhaysingh7/IBM-Capstone-Project/blob/main/jupyter-labs-webscraping.ipynb>

```
1. Apply HTTP Get method to request the Falcon 9 rocket launch page

In [4]: static_url = "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922"

In [5]: # use requests.get() method with the provided static_url
        # assign the response to a object
        html_data = requests.get(static_url)
        html_data.status_code

Out[5]: 200

2. Create a BeautifulSoup object from the HTML response

In [6]: # Use BeautifulSoup() to create a BeautifulSoup object from a response text content
        soup = BeautifulSoup(html_data.text, 'html.parser')

        Print the page title to verify if the BeautifulSoup object was created properly

In [7]: # Use soup.title attribute
        soup.title

Out[7]: <title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>

3. Extract all column names from the HTML table header

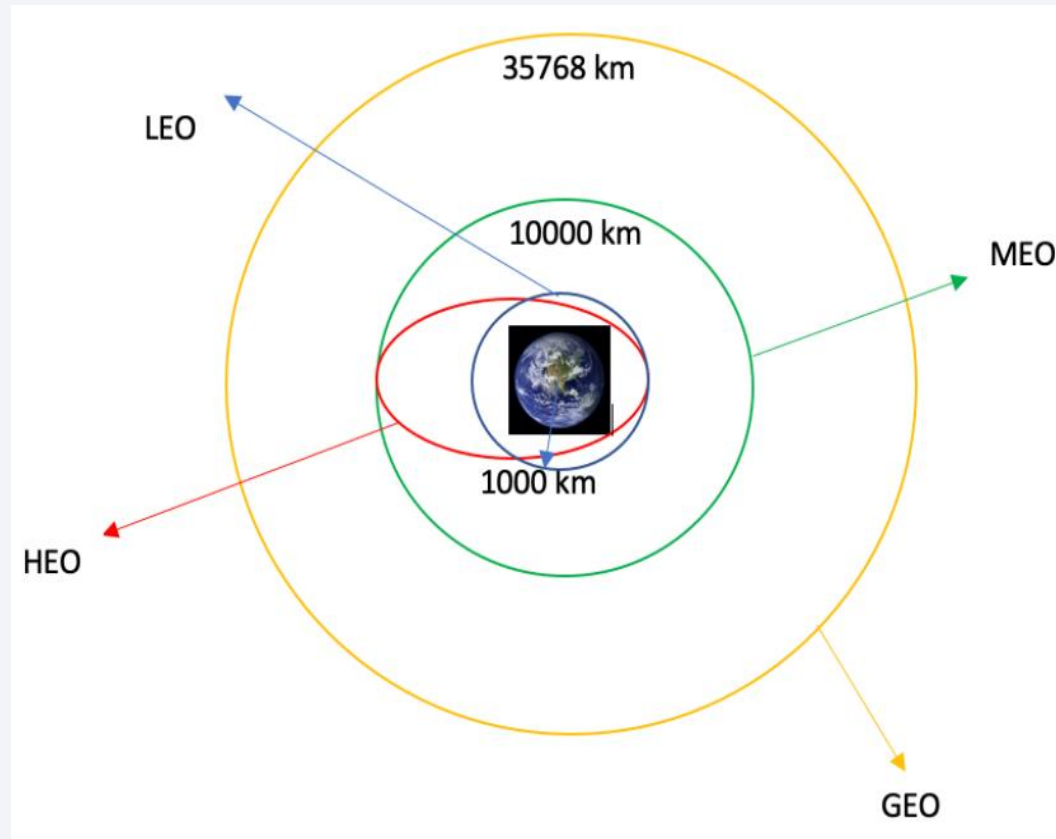
In [10]: column_names = []

        # Apply find_all() function with 'th' element on first_launch_table
        # Iterate each th element and apply the provided extract_column_from_header() to get a column name
        # Append the Non-empty column name ('if name is not None and len(name) > 0') into a list called column_names

        element = soup.find_all('th')
        for row in range(len(element)):
            try:
                name = extract_column_from_header(element[row])
                if (name is not None and len(name) > 0):
                    column_names.append(name)
            except:
                pass

4. Create a dataframe by parsing the launch HTML tables
5. Export data to csv
```

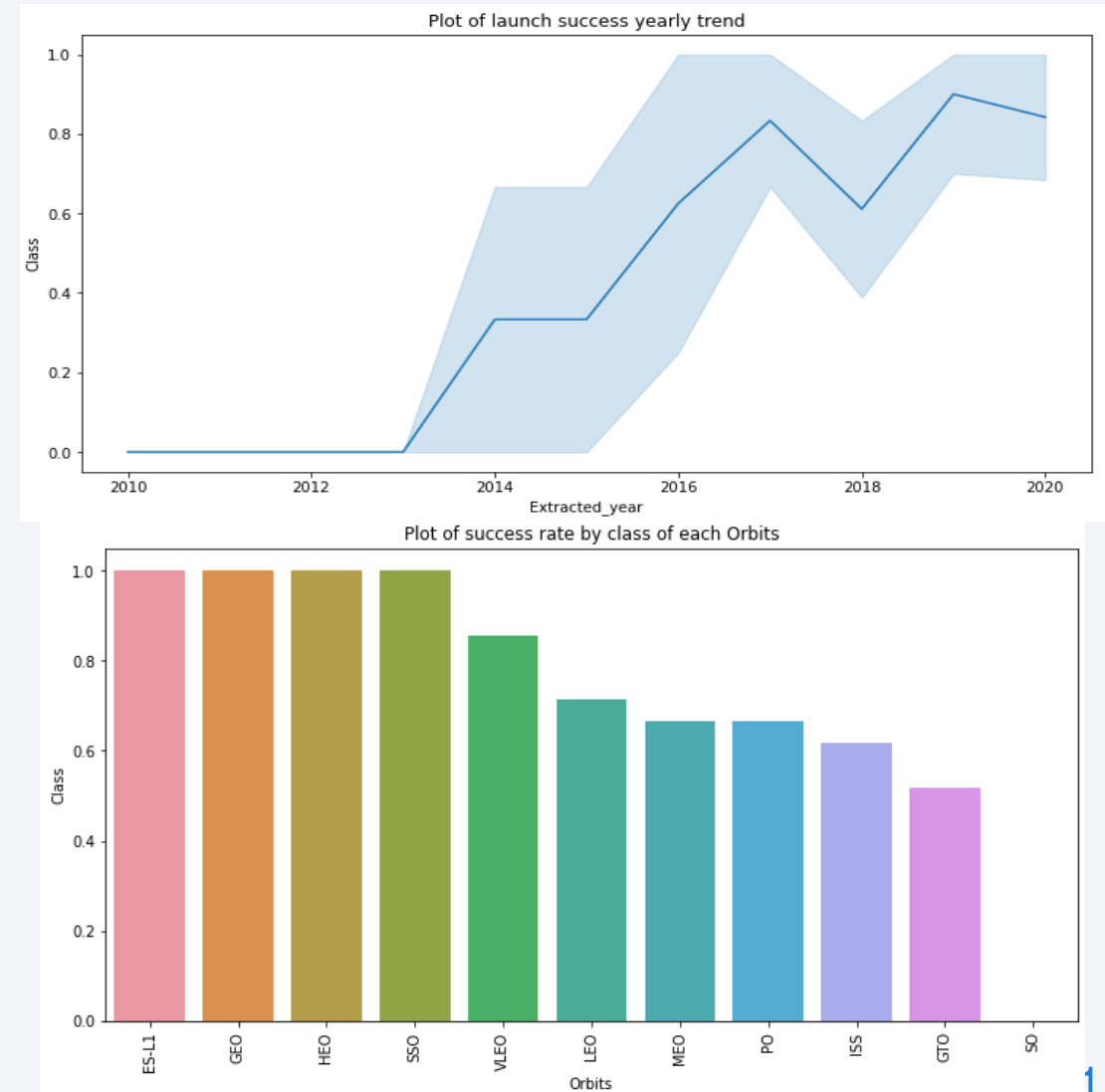
Data Wrangling



- Cleaned and preprocessed the Falcon 9 launch dataset by handling missing values and selecting relevant features.
- Performed exploratory analysis to identify launch sites, orbit types, and landing outcomes.
- Created a binary **landing success label** from the mission outcome for classification.
- Exported the cleaned dataset as a **CSV file** for visualization and machine learning.
- **Notebook:**
GitHub:
<https://github.com/abhaysingh7/IBM-Capstone-Project/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>

EDA with Data Visualization

- Explored relationships between **flight number, launch site, payload mass, orbit type, and landing success** using Python visualizations.
- Analyzed launch success trends across different launch sites, orbit types, and mission years.
- Identified key factors influencing Falcon 9 first-stage landing success through scatter plots, bar charts, and trend analysis.
- Used the insights to guide feature selection for the machine learning models.
- **Notebook:**
GitHub:
<https://github.com/abhaysingh7/IBM-Capstone-Project/blob/main/edadataviz.ipynb>



EDA with SQL

- Loaded the cleaned SpaceX launch dataset into a **PostgreSQL** database for SQL-based analysis.
- Wrote SQL queries to analyze **launch sites, payload mass, booster versions, and mission outcomes**.
- Retrieved key insights, including:
 - Unique Falcon 9 launch sites.
 - Total and average payload mass by booster version.
 - Successful and failed mission outcomes.
 - Failed drone-ship landings by booster version and launch site.
- Used SQL insights to validate trends identified during exploratory data analysis.
- **Notebook:**
GitHub: https://github.com/abhaysingh7/IBM-Capstone-Project/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

- We marked all launch sites, and added map objects such as markers, circles, lines to mark the success or failure of launches for each site on the folium map.
- We assigned the feature launch outcomes (failure or success) to class 0 and 1.i.e., 0 for failure, and 1 for success.
- Using the color-labeled marker clusters, we identified which launch sites have relatively high success rate.
- We calculated the distances between a launch site to its proximities. We answered some question for instance:
 - Are launch sites near railways, highways and coastlines.
 - Do launch sites keep certain distance away from cities.

Build a Dashboard with Plotly Dash

- Developed an interactive **Plotly Dash dashboard** to explore Falcon 9 launch data.
- Created **pie charts** to visualize the distribution of successful launches across different launch sites.
- Built **scatter plots** to analyze the relationship between **payload mass** and **landing success** for different booster versions.
- Added interactive filters to compare launch outcomes by **launch site** and **payload range**, enabling dynamic data exploration.
- **Notebook:**
GitHub: https://github.com/abhaysingh7/IBM-Capstone-Project/blob/main/spacex_app.py

Predictive Analysis (Classification)

- Preprocessed the dataset using **NumPy** and **Pandas**, and split it into **training** and **testing** sets.
- Trained and optimized **Logistic Regression, SVM, Decision Tree, and KNN** models using **GridSearchCV**.
- Evaluated model performance using **accuracy** and compared the results across all classifiers.
- Applied **feature engineering** and **hyperparameter tuning** to improve predictive performance.
- Selected the **best-performing classification model** for predicting Falcon 9 first-stage landing success.
- **Notebook:**
GitHub: <https://github.com/abhaysingh7/IBM-Capstone-Project/blob/main/Machine%20Learning%20Prediction.ipynb>

Results

Exploratory Data Analysis (EDA)

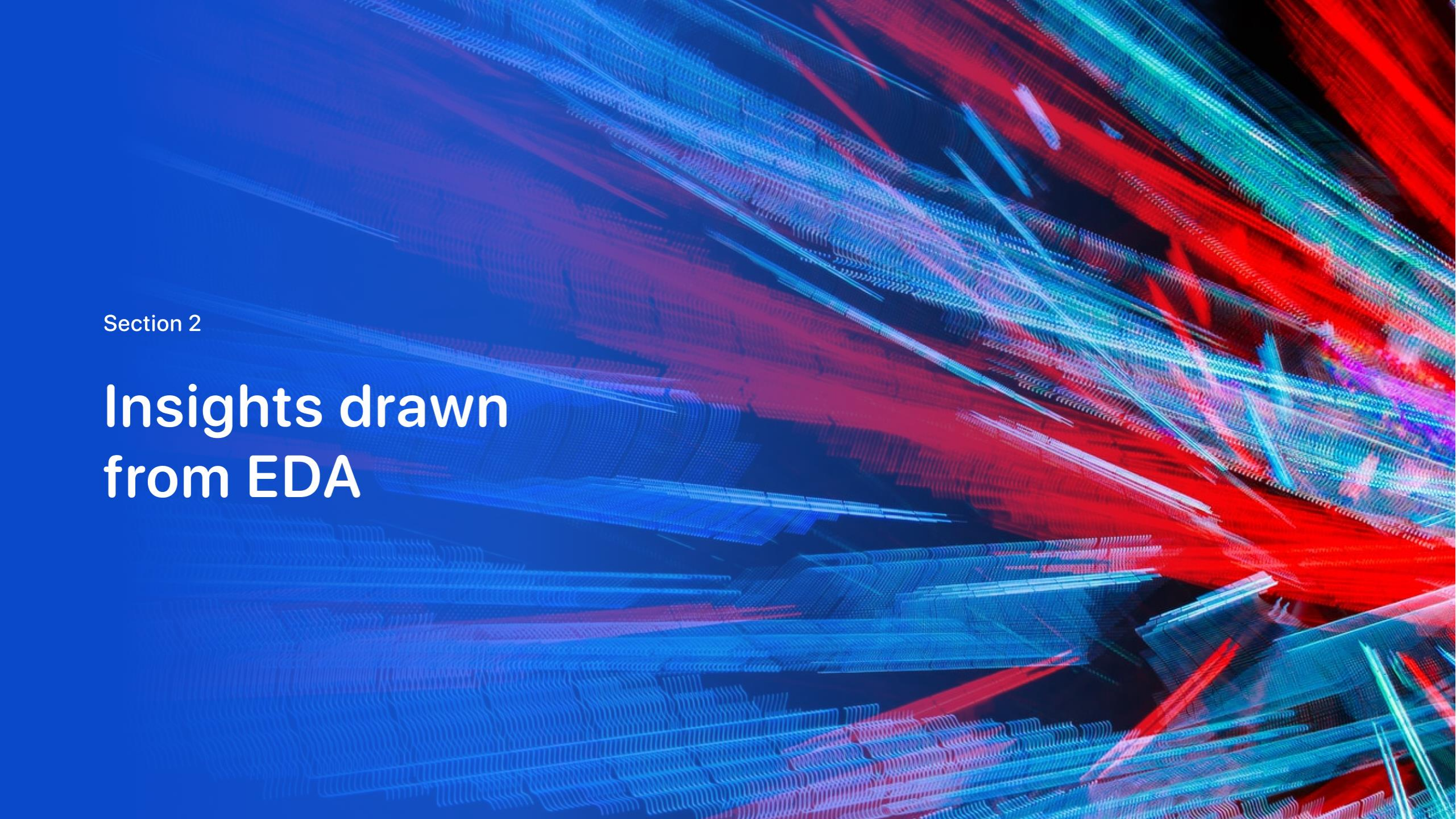
- Launch success increased over time with higher flight numbers.
- Launch site, payload mass, and orbit type significantly influenced landing success.
- SQL analysis validated trends in launch frequency, payload distribution, and mission outcomes.

Interactive Visual Analytics

- Folium maps identified launch sites with higher landing success rates and their proximity to geographical features.
- Plotly Dash dashboard enabled interactive exploration of launch success by launch site, payload mass, and booster version.

Predictive Analysis

- Compared **Logistic Regression, SVM, Decision Tree, and KNN** classification models.
- **Decision Tree** achieved the highest prediction accuracy and was selected as the final model for predicting Falcon 9 first-stage landing success.

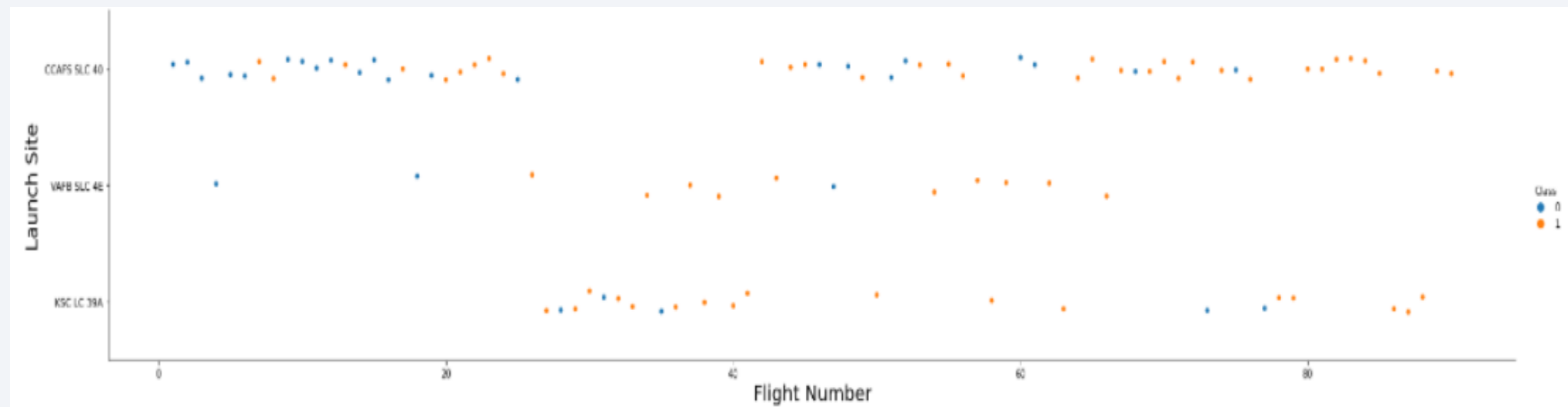
The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a dynamic pattern of diagonal streaks in shades of blue and red on the right. These streaks are layered over a fine, light-colored grid, creating a sense of depth and movement, reminiscent of a digital or data visualization theme.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

- The analysis shows that **launch success rates generally increased with flight number** across all launch sites.
- Launch sites with a **higher number of Falcon 9 missions** demonstrated better landing success, indicating improved operational reliability over time.
- **KSC LC-39A** and **CCSFS SLC-40** achieved the highest success rates due to increased launch frequency and booster reuse experience.
- **Key Insight:** The results suggest that **operational experience and continuous technological improvements** contributed significantly to the successful recovery of the Falcon 9 first-stage booster.
- **Takeaway:** *Higher launch frequency is associated with improved landing success, highlighting the impact of experience and iterative engineering improvements on Falcon 9 missions.*



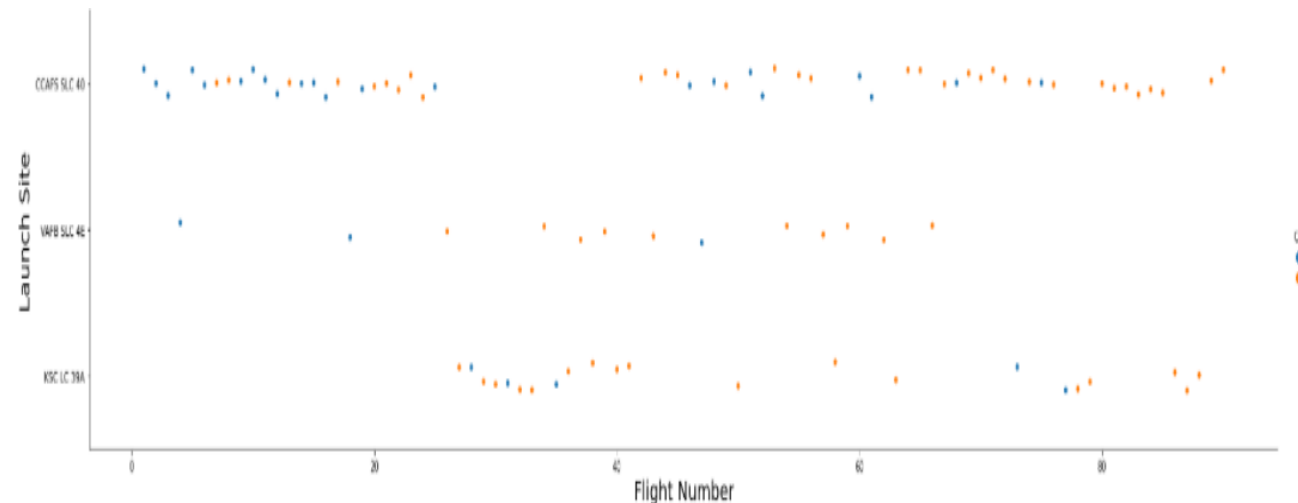
Payload vs. Launch Site

- Payload mass varied across different Falcon 9 launch sites, with each site supporting different mission profiles.
- Successful landings were observed across a wide range of payload masses, although heavier payloads generally posed greater landing challenges.
- **KSC LC-39A** and **CCSFS SLC-40** handled a larger number of launches and demonstrated consistently higher landing success rates.
- **Key Insight:** While **payload mass influences landing success, launch site characteristics and operational experience** also play an important role in achieving successful first-stage recoveries.

Takeaway: *Landing success depends on multiple factors, with payload mass and launch site jointly influencing Falcon 9 booster recovery performance*

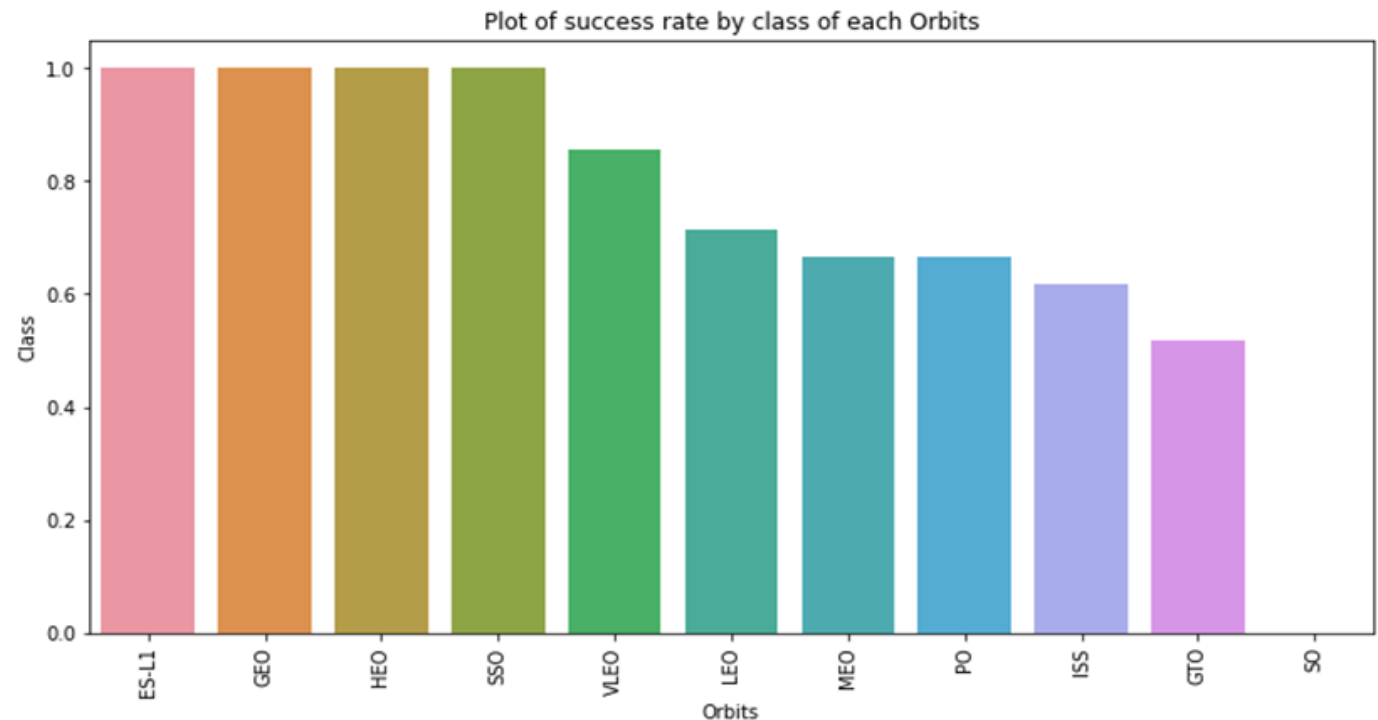


The greater the payload mass for launch site CCAFS SLC 40 the higher the success rate for the rocket.



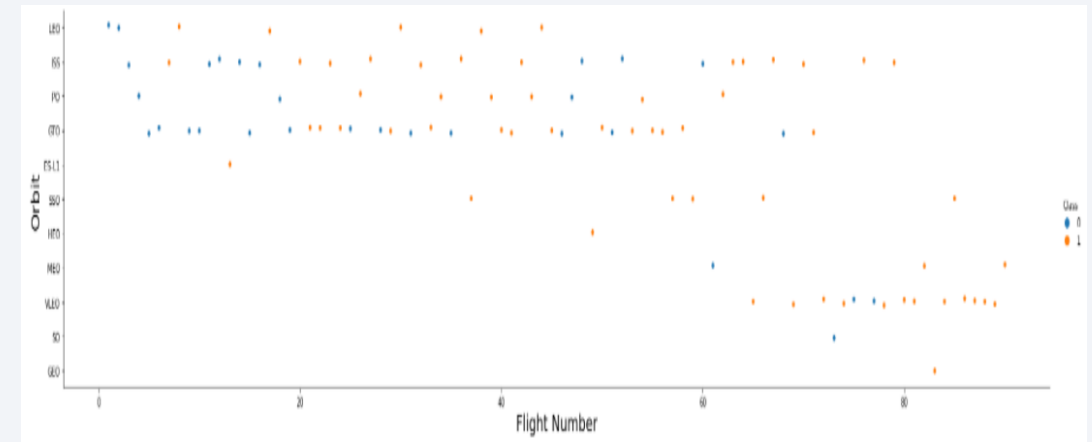
Success Rate vs. Orbit Type

- Landing success rates varied across different orbit types, indicating that **mission profile influences booster recovery**.
- **ES-L1, GEO, HEO, SSO, and VLEO** missions achieved the **highest landing success rates** among the observed orbit categories.
- Lower success rates in some orbit types may be attributed to **higher mission complexity, greater energy requirements, or operational constraints**.
- **Key Insight:** Orbit type is an important feature for predicting Falcon 9 first-stage landing success and should be included in the machine learning model.
- **Takeaway:** *Orbit type significantly affects the probability of a successful first-stage landing, making it one of the key predictors in the classification model.*



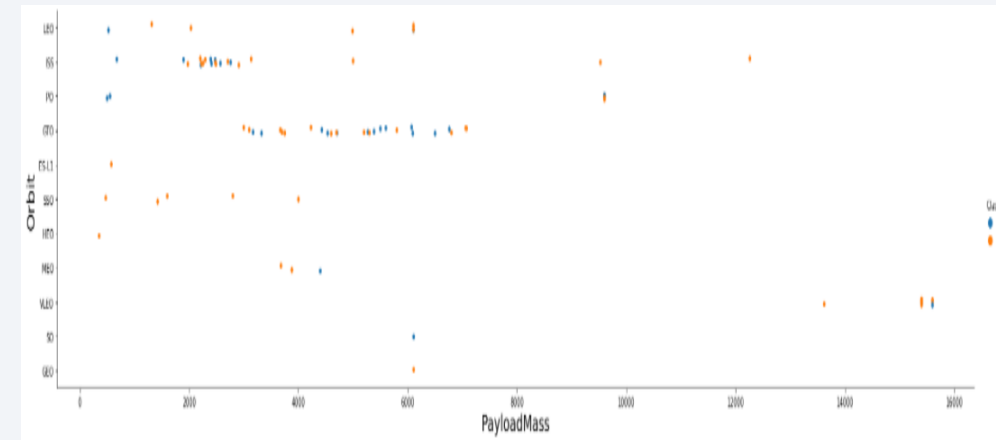
Flight Number vs. Orbit Type

- The relationship between **flight number** and **landing success** varies across different orbit types.
- For **LEO (Low Earth Orbit)** missions, landing success generally **improves as the flight number increases**, indicating that increased operational experience contributes to higher booster recovery rates.
- In contrast, **GTO (Geostationary Transfer Orbit)** missions show **no clear relationship** between flight number and landing success, likely due to their higher mission complexity and energy requirements.
- **Key Insight:** Both **flight experience** and **orbit type** are important predictors of Falcon 9 first-stage landing success and should be included in the machine learning model.
- **Takeaway:** *Operational experience improves landing success for LEO missions, while GTO missions remain more challenging regardless of flight number.*



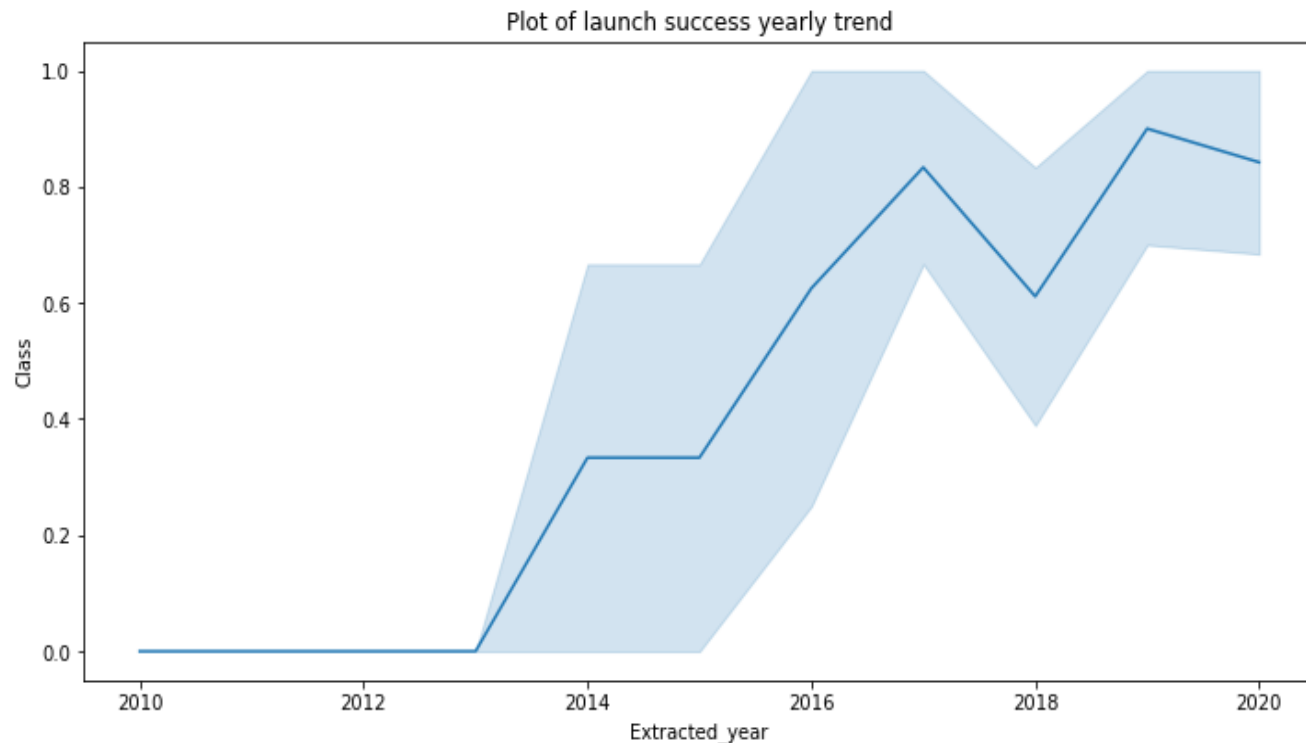
Payload vs. Orbit Type

- The relationship between **payload mass** and **landing success** varies across different orbit types.
- Missions carrying **heavier payloads** achieved higher landing success rates in **PO, LEO, and ISS** orbits.
- For other orbit types, successful landings were less frequent, suggesting that **mission profile and orbit characteristics** influence booster recovery performance.
- **Key Insight:** Payload mass alone does not determine landing success; its impact depends on the **target orbit** and overall mission requirements.
- **Takeaway:** *Payload mass and orbit type together play a significant role in predicting Falcon 9 first-stage landing success and are valuable features for the classification model.*



Launch Success Yearly Trend

- The analysis shows that the **Falcon 9 first-stage landing success rate increased steadily from 2013 to 2020.**
- The upward trend reflects **continuous improvements in booster technology, operational experience, and landing procedures.**
- By **2020**, SpaceX achieved consistently high landing success rates, demonstrating the effectiveness of its reusable rocket program.
- **Key Insight:** Launch year is an important predictor of landing success, indicating that technological advancements and accumulated experience significantly improved Falcon 9 reliability over time.
- **Takeaway:** *The steady increase in landing success from 2013–2020 demonstrates the positive impact of continuous innovation and operational learning on Falcon 9 mission performance.*



All Launch Site Names

- Used the **DISTINCT** keyword to retrieve the unique Falcon 9 launch sites from the SpaceX dataset.
- The query identified all launch locations used for Falcon 9 missions, eliminating duplicate records.
- The results provide an overview of the launch infrastructure and support comparisons of launch frequency and landing success across sites.
- **Key Insight:** Identifying unique launch sites is essential for analyzing how launch location influences Falcon 9 first-stage landing success.
- **Takeaway:** *Launch site is a key categorical feature that contributes to differences in mission outcomes and landing success rates.*

Display the names of the unique launch sites in the space mission

```
In [10]: task_1 = '''
          SELECT DISTINCT LaunchSite
          FROM SpaceX
          ...
          create_pandas_df(task_1, database=conn)
```

```
Out[10]:
```

	launchsite
0	KSC LC-39A
1	CCAFS LC-40
2	CCAFS SLC-40
3	VAFB SLC-4E

Launch Site Names Begin with 'CCA'

- Used the SQL **LIKE** 'CCA%' operator to retrieve launch sites whose names begin with "CCA".
- The query filtered the dataset to display **Cape Canaveral launch sites**, making it easier to analyze missions from a specific location.
- The results help compare launch activity and landing performance between Cape Canaveral and other SpaceX launch sites.
- **Key Insight:** SQL filtering enables targeted analysis of launch sites, allowing more detailed exploration of mission outcomes and operational performance.

Takeaway: *Filtering data by launch site helps identify location-specific trends that may influence Falcon 9 landing success.*

Display 5 records where launch sites begin with the string 'CCA'

```
In [11]: task_2 = '''
        SELECT *
        FROM SpaceX
        WHERE LaunchSite LIKE 'CCA%'
        LIMIT 5
        '''

        create_pandas_df(task_2, database=conn)
```

```
Out[11]:
```

	date	time	boosterversion	launchsite	payload	payloadmasskg	orbit	customer	missionoutcome	landingoutcome
0	2010-04-06	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
1	2010-08-12	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of...	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2	2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
3	2012-08-10	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
4	2013-01-03	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Used an SQL **SUM()** query to calculate the total payload mass carried by **NASA (CRS)** missions.
- The analysis showed that **NASA (CRS) boosters transported a total payload of 45,596 kg.**
- This result highlights the significant contribution of Falcon 9 to NASA's Commercial Resupply Services (CRS) missions.
- **Key Insight:** SQL aggregation functions provide valuable insights into payload capacity and mission performance across different customers and mission types.
- **Takeaway:** *Falcon 9 successfully delivered a cumulative payload of **45,596 kg** for NASA (CRS) missions, demonstrating its capability for large-scale cargo transportation.*

Display the total payload mass carried by boosters launched by NASA (CRS)

```
In [12]: task_3 = '''
          SELECT SUM(PayloadMassKG) AS Total_PayloadMass
          FROM SpaceX
          WHERE Customer LIKE 'NASA (CRS)'
          '''
          create_pandas_df(task_3, database=conn)
```

```
Out[12]:
```

	total_payloadmass
0	45596

Average Payload Mass by F9 v1.1

- Used the SQL **AVG()** function to calculate the average payload mass carried by the **Falcon 9 v1.1** booster.
- The analysis showed that the **average payload mass was 2,928.4 kg**.
- This metric provides insight into the typical carrying capacity and operational performance of the **F9 v1.1** booster version.
- **Key Insight:** Average payload analysis helps compare the performance of different booster versions and supports the evaluation of mission capabilities.
- **Takeaway:** *The Falcon 9 v1.1 booster carried an average payload of 2,928.4 kg, demonstrating its effectiveness in supporting a wide range of commercial and government missions.*

Display average payload mass carried by booster version F9 v1.1

```
In [13]: task_4 = '''
          SELECT AVG(PayloadMassKG) AS Avg_PayloadMass
          FROM SpaceX
          WHERE BoosterVersion = 'F9 v1.1'
          '''
          create_pandas_df(task_4, database=conn)
```

```
Out[13]:
```

	avg_payloadmass
0	2928.4

First Successful Ground Landing Date

- Used an SQL query to identify the **first successful Falcon 9 landing on a ground landing pad**.
- The analysis showed that the **first successful ground landing occurred on 22 December 2015**.
- This milestone marked a significant advancement in **SpaceX's reusable rocket program**, demonstrating the feasibility of recovering and reusing first-stage boosters.
- **Key Insight:** Successful ground landings represented a major breakthrough that contributed to reducing launch costs and improving mission sustainability.
- **Takeaway:** *The first successful ground landing on **22 December 2015** was a pivotal achievement that validated SpaceX's reusable launch vehicle technology.*

In [14]:

```
task_5 = '''
    SELECT MIN(Date) AS FirstSuccessfull_landing_date
    FROM SpaceX
    WHERE LandingOutcome LIKE 'Success (ground pad)'
    '''

create_pandas_df(task_5, database=conn)
```

Out[14]:

	firstsuccessfull_landing_date
--	-------------------------------

0	2015-12-22
---	------------

Successful Drone Ship Landing with Payload between 4000 and 6000

```
In [15]: task_6 = '''
          SELECT BoosterVersion
          FROM SpaceX
          WHERE LandingOutcome = 'Success (drone ship)'
             AND PayloadMassKG > 4000
             AND PayloadMassKG < 6000
          ...
          create_pandas_df(task_6, database=conn)
```

```
Out[15]:
```

	boosterversion
0	F9 FT B1022
1	F9 FT B1026
2	F9 FT B1021.2
3	F9 FT B1031.2

- Used the SQL **WHERE** clause with **AND** conditions to filter missions that **successfully landed on a drone ship** with a **payload mass between 4,000 kg and 6,000 kg**.
- The query identified missions that met both the landing success and payload criteria.
- The results demonstrate that Falcon 9 successfully recovered first-stage boosters while carrying **medium-to-heavy payloads**.
- **Key Insight:** SpaceX consistently achieved successful drone ship recoveries for missions within this payload range, highlighting the reliability of its reusable booster technology.
- **Takeaway:** *Successful booster recovery is achievable even for missions carrying **4,000–6,000 kg payloads**, demonstrating Falcon 9's capability to support demanding commercial and government launches.*

Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

```
In [16]: task_7a = '''
          SELECT COUNT(MissionOutcome) AS SuccessOutcome
          FROM SpaceX
          WHERE MissionOutcome LIKE 'Success%'
          '''

          task_7b = '''
          SELECT COUNT(MissionOutcome) AS FailureOutcome
          FROM SpaceX
          WHERE MissionOutcome LIKE 'Failure%'
          '''

          print('The total number of successful mission outcome is:')
          display(create_pandas_df(task_7a, database=conn))
          print()
          print('The total number of failed mission outcome is:')
          create_pandas_df(task_7b, database=conn)
```

The total number of successful mission outcome is:

	successoutcome
0	100

The total number of failed mission outcome is:

```
Out[16]: failureoutcome
0         1
```

- Used the SQL **LIKE** operator with wildcard (%) to filter mission outcomes classified as **successful** or **failed**.
- Calculated the total number of successful and unsuccessful Falcon 9 landing attempts from the dataset.
- The results provide an overview of Falcon 9 landing performance and mission reliability.
- **Key Insight:** The analysis shows that successful landings outnumbered failures, reflecting the continuous improvement of SpaceX's reusable rocket technology over time.
- **Takeaway:** *Comparing successful and failed mission outcomes helps evaluate Falcon 9's operational reliability and supports the development of accurate landing prediction models.*

Boosters Carried Maximum Payload

- Used the SQL **MAX()** function together with a **subquery** to identify the booster(s) that carried the highest payload.
- The query retrieved the booster version and mission associated with the **maximum payload mass** in the dataset.
- This analysis highlights the lifting capability of Falcon 9 boosters for high-payload missions.
- **Key Insight:** Identifying the maximum payload helps evaluate booster performance and compare the capabilities of different Falcon 9 configurations.
- **Takeaway:** *The booster carrying the maximum payload demonstrates Falcon 9's ability to support demanding commercial and government space missions while maintaining successful launch operations.*

List the names of the booster_versions which have carried the maximum payload mass. Use a subquery

In [17]:

```
task_8 = '''
SELECT BoosterVersion, PayloadMassKG
FROM SpaceX
WHERE PayloadMassKG = (
    SELECT MAX(PayloadMassKG)
    FROM SpaceX
)
ORDER BY BoosterVersion
'''
create_pandas_df(task_8, database=conn)
```

Out[17]:

	boosterversion	payloadmasskg
0	F9 B5 B1048.4	15600
1	F9 B5 B1048.5	15600
2	F9 B5 B1049.4	15600
3	F9 B5 B1049.5	15600
4	F9 B5 B1049.7	15600
5	F9 B5 B1051.3	15600
6	F9 B5 B1051.4	15600
7	F9 B5 B1051.6	15600
8	F9 B5 B1056.4	15600
9	F9 B5 B1058.3	15600
10	F9 B5 B1060.2	15600
11	F9 B5 B1060.3	15600

2015 Launch Records

- We used a combinations of the **WHERE** clause, **LIKE**, **AND**, and **BETWEEN** conditions to filter for failed landing outcomes in drone ship, their booster versions, and launch site names for year 2015

```
List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

In [18]: task_9 = '''
          SELECT BoosterVersion, LaunchSite, LandingOutcome
          FROM SpaceX
          WHERE LandingOutcome LIKE 'Failure (drone ship)'
             AND Date BETWEEN '2015-01-01' AND '2015-12-31'
          ...
          create_pandas_df(task_9, database=conn)

Out[18]:
```

	boosterversion	launchsite	landingoutcome
0	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
1	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad))

```
In [19]: task_10 = '''
          SELECT LandingOutcome, COUNT(LandingOutcome)
          FROM SpaceX
          WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20'
          GROUP BY LandingOutcome
          ORDER BY COUNT(LandingOutcome) DESC
          '''
          create_pandas_df(task_10, database=conn)
```

```
Out[19]:
```

	landingoutcome	count
0	No attempt	10
1	Success (drone ship)	6
2	Failure (drone ship)	5
3	Success (ground pad)	5
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Precluded (drone ship)	1
7	Failure (parachute)	1

- We selected Landing outcomes and the **COUNT** of landing outcomes from the data and used the **WHERE** clause to filter for landing outcomes **BETWEEN** 2010-06-04 to 2017-03-20.
- We applied the **GROUP BY** clause to group the landing outcomes and the **ORDER BY** clause to order the grouped landing outcome in descending order.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a thin, curved line separating the dark surface from the deep blue of space.

Section 3

Launch Sites Proximities Analysis

All launch sites global map markers

- Created an interactive **Folium map** to visualize the geographical locations of all Falcon 9 launch sites.
- Added **map markers** to identify each launch site and enable interactive exploration.
- The visualization provides a clear overview of the spatial distribution of SpaceX launch facilities.
- **Key Insight:** Most launch sites are located along the **U.S. coastline**, providing safe launch trajectories over the ocean and facilitating first-stage booster recovery operations.

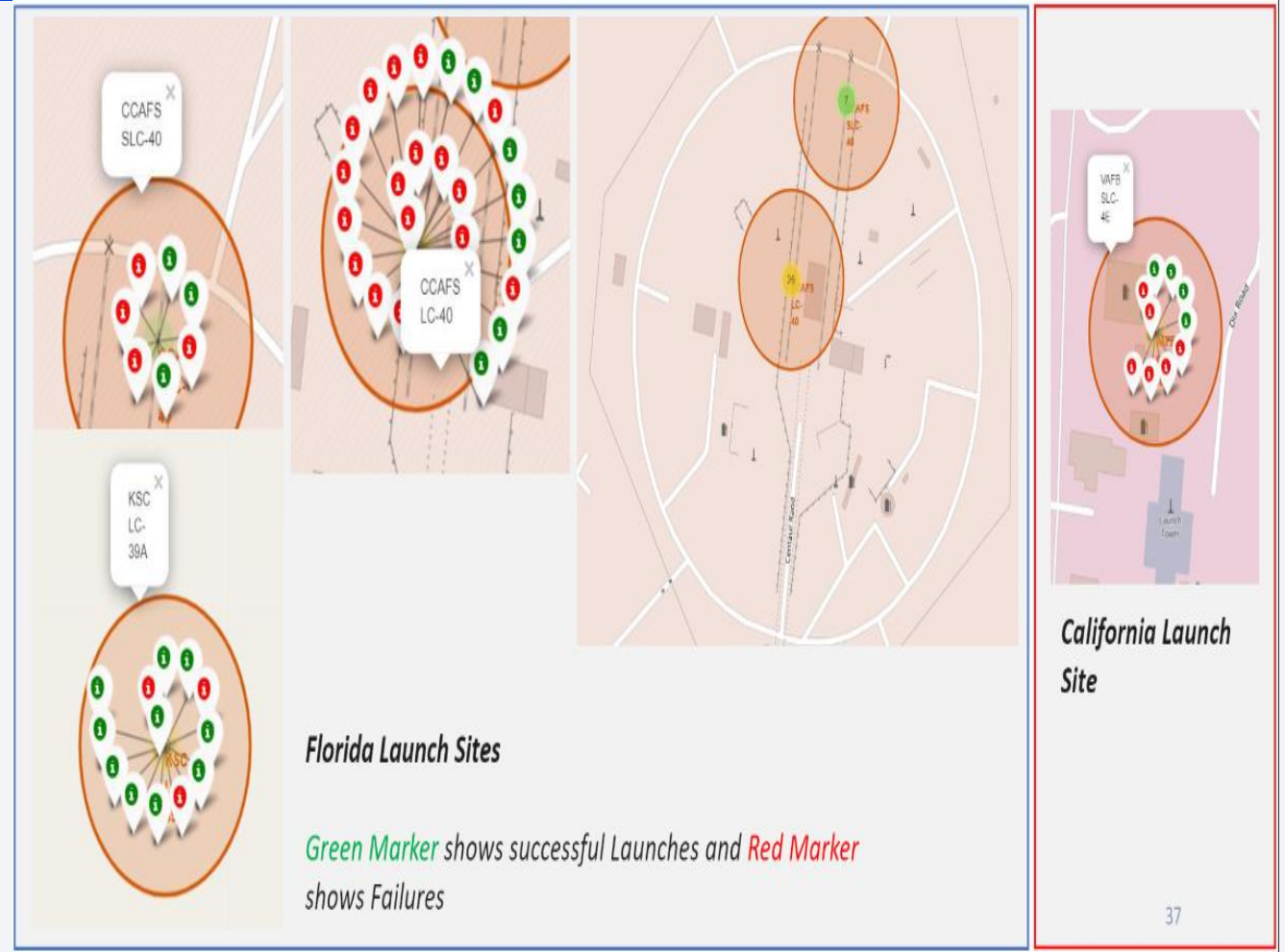
Takeaway: Mapping launch sites geographically helps understand how location influences launch operations, mission planning, and booster recovery success.



Markers showing launch sites with color labels

- Added **color-coded markers** to distinguish **successful (green)** and **unsuccessful (red)** Falcon 9 first-stage landings.
- Used **marker clusters** to improve map readability and visualize multiple launch records at each site.
- The color distribution provides an immediate comparison of landing success rates across different launch sites.
- **Key Insight: KSC LC-39A and CCSFS SLC-40** display a higher concentration of successful landings, indicating stronger operational performance.

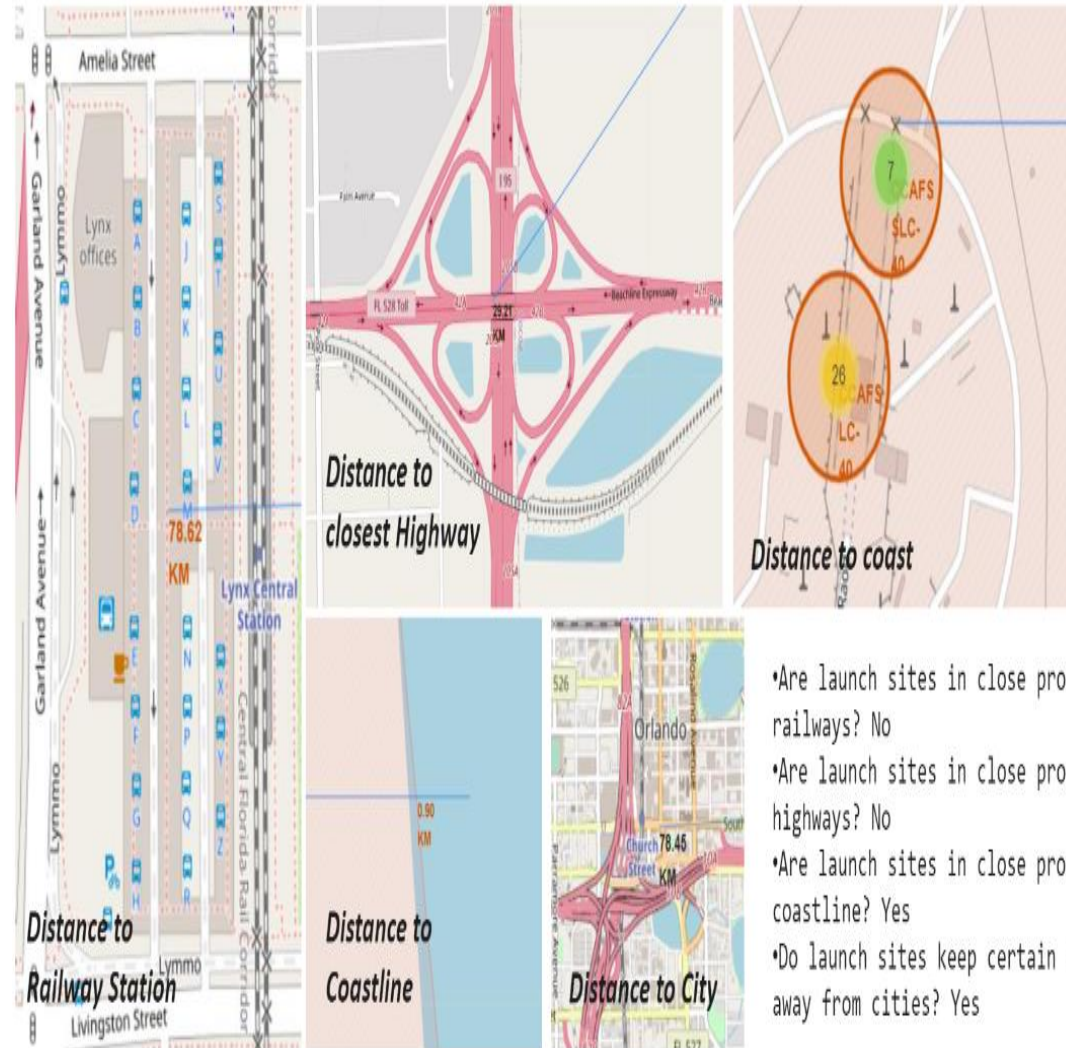
Takeaway: *Color-labeled markers enable quick identification of launch sites with higher landing success and support the analysis of geographical patterns in Falcon 9 missions.*



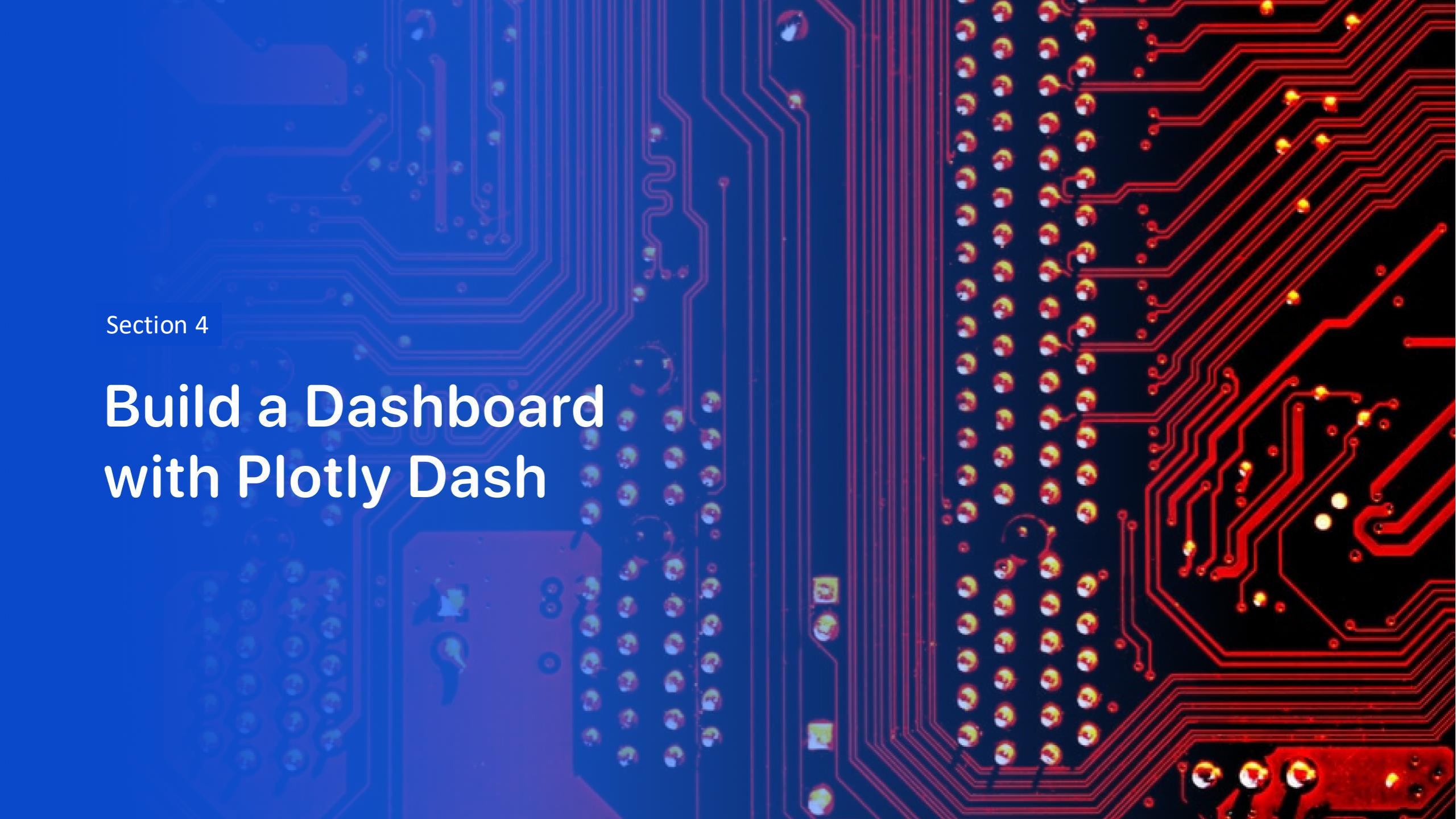
Launch Site distance to landmarks

- Calculated the distances between each **Falcon 9 launch site** and nearby **coastlines, highways, railways, and cities** using Folium.
- Visualized these distances with map markers and connecting lines to examine the geographical surroundings of each launch site.
- The analysis showed that launch sites are located **close to coastlines** while maintaining a **safe distance from densely populated urban areas**.
- **Key Insight:** The strategic placement of launch sites enhances operational safety, simplifies logistics, and supports efficient first-stage booster recovery.

Takeaway: *The proximity of launch sites to key geographical features plays an important role in mission planning, launch safety, and successful Falcon 9 recovery operations.*



- Are launch sites in close proximity to railways? No
- Are launch sites in close proximity to highways? No
- Are launch sites in close proximity to coastline? Yes
- Do launch sites keep certain distance away from cities? Yes



Section 4

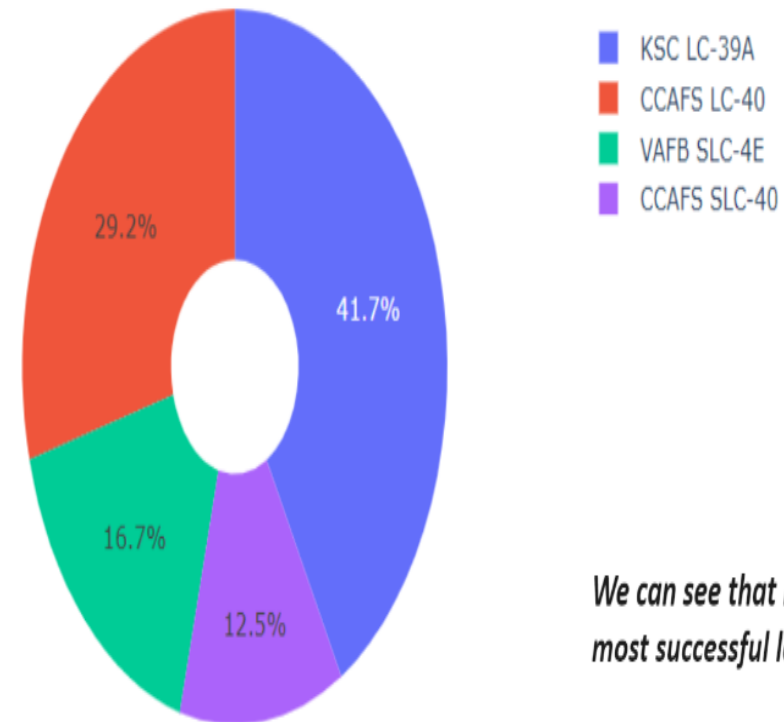
Build a Dashboard with Plotly Dash

Pie chart showing the success percentage achieved by each launch site

- Developed an interactive **Plotly Dash pie chart** to visualize the distribution of successful launches across Falcon 9 launch sites.
- The chart compares the percentage of successful missions for each launch location.
- The results show that **KSC LC-39A** and **CCSFS SLC-40** achieved the highest launch success rates, reflecting consistent operational performance.
- **Key Insight:** Launch site plays a significant role in mission success and is an important feature for predicting Falcon 9 first-stage landing outcomes.

Takeaway: *The pie chart provides a clear comparison of launch site performance, highlighting the locations with the highest Falcon 9 landing success rates.*

Total Success Launches By all sites

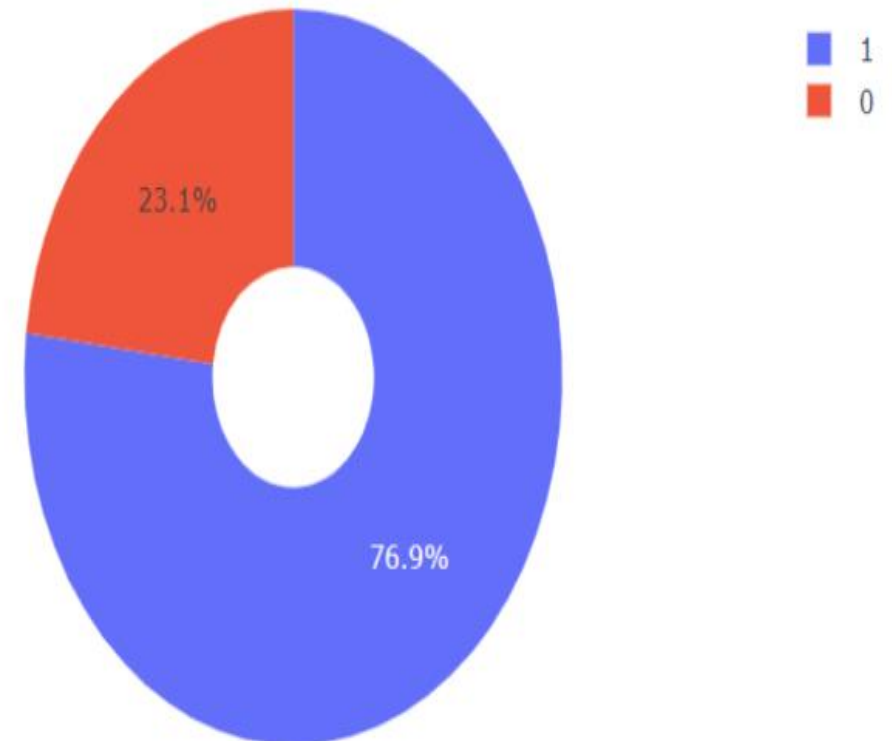


We can see that KSC LC-39A had the most successful launches from all the sites

Pie chart showing the Launch site with the highest launch success ratio

- Developed an interactive **Plotly Dash pie chart** to compare the launch success ratios across all Falcon 9 launch sites.
- The visualization identifies the launch site with the **highest proportion of successful first-stage landings**.
- The results indicate that **KSC LC-39A** achieved the highest launch success ratio, demonstrating consistently reliable mission performance.
- **Key Insight:** Launch site is a significant factor influencing landing success and should be included as a predictive feature in the machine learning model.

Takeaway: *Comparing launch success ratios across sites highlights operational differences and identifies the most reliable Falcon 9 launch location.*

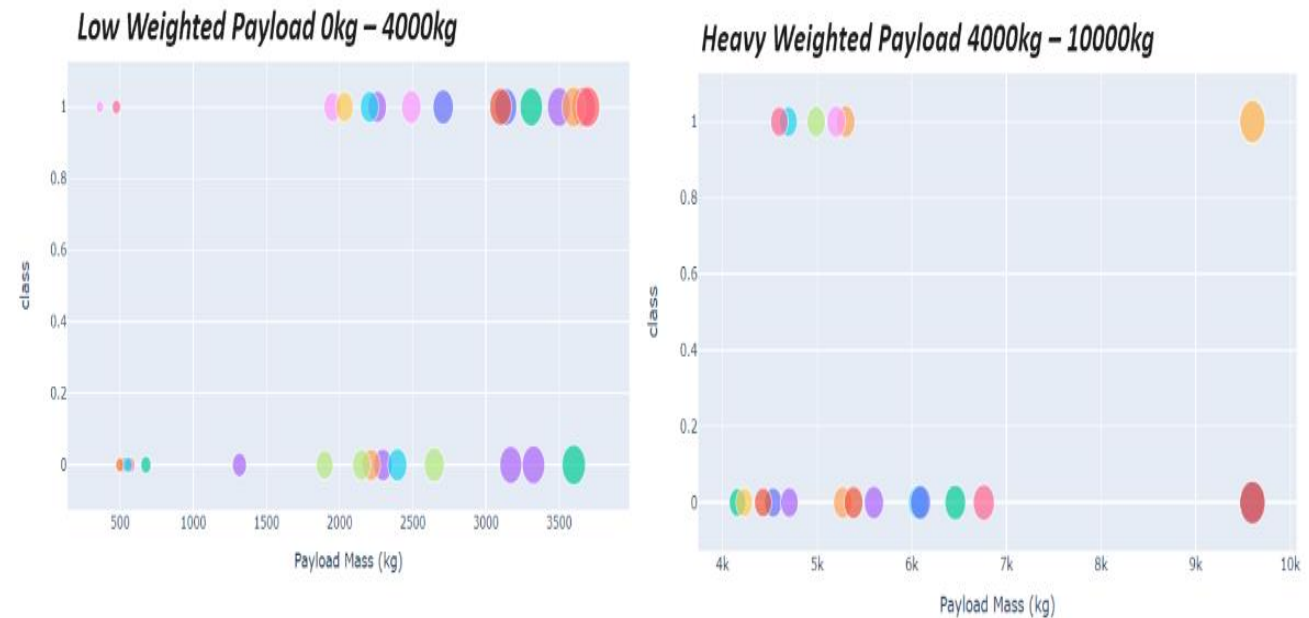


KSC LC-39A achieved a 76.9% success rate while getting a 23.1% failure rate

Scatter plot of Payload vs Launch Outcome for all sites, with different payload selected in the range slider

- Developed an interactive **Plotly Dash** scatter plot to visualize the relationship between **payload mass** and **launch outcome** across all Falcon 9 launch sites.
- Incorporated a **payload range slider** to dynamically filter missions and analyze landing success for different payload ranges.
- The visualization shows that **successful landings occur across a wide range of payload masses**, although the success rate varies depending on mission characteristics and launch site.
- **Key Insight:** Payload mass influences landing success, but it is **not the sole determining factor**. Other variables such as **orbit type, launch site, and booster version** also significantly affect first-stage recovery.

Takeaway: *The interactive scatter plot demonstrates that landing success is driven by multiple factors, making a machine learning approach appropriate for predicting Falcon 9 first-stage landings.*



We can see the success rates for low weighted payloads is higher than the heavy weighted payloads

Section 5

Predictive Analysis (Classification)

Classification Accuracy

- The decision tree classifier is the model with the highest classification accuracy

- Evaluated multiple classification models, including **Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN)**.
- Applied **GridSearchCV** to optimize model hyperparameters and improve predictive performance.
- Compared the models using **classification accuracy** on the test dataset.
- **Decision Tree** achieved the **highest accuracy (94.4%)**, making it the best-performing model for predicting Falcon 9 first-stage landing success.
- **Key Insight:** Model tuning and feature engineering significantly improved prediction accuracy, demonstrating that machine learning can effectively predict landing outcomes.

Takeaway: *The Decision Tree classifier provided the most accurate predictions and was selected as the final model for this project.*

```
models = {'KNeighbors': knn_cv.best_score_,
          'DecisionTree': tree_cv.best_score_,
          'LogisticRegression': logreg_cv.best_score_,
          'SupportVector': svm_cv.best_score_}

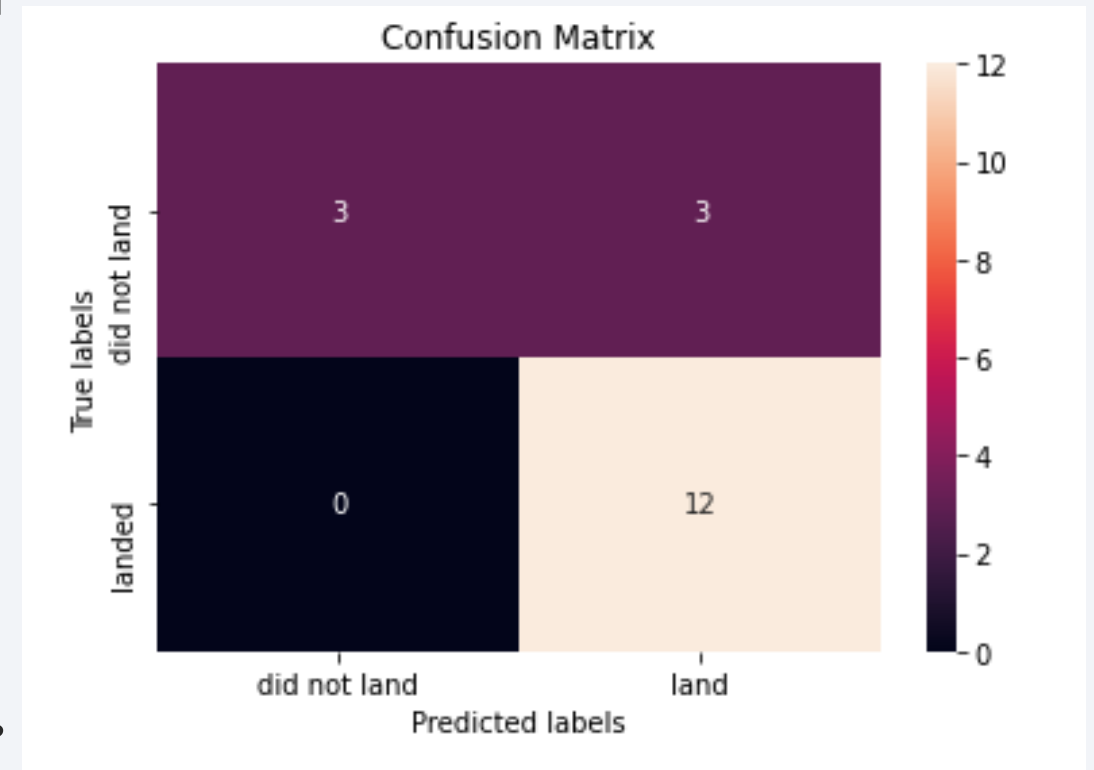
bestalgorithm = max(models, key=models.get)
print('Best model is', bestalgorithm, 'with a score of', models[bestalgorithm])
if bestalgorithm == 'DecisionTree':
    print('Best params is:', tree_cv.best_params_)
if bestalgorithm == 'KNeighbors':
    print('Best params is:', knn_cv.best_params_)
if bestalgorithm == 'LogisticRegression':
    print('Best params is:', logreg_cv.best_params_)
if bestalgorithm == 'SupportVector':
    print('Best params is:', svm_cv.best_params_)
```

Best model is DecisionTree with a score of 0.8732142857142856

Best params is : {'criterion': 'gini', 'max_depth': 6, 'max_features': 'auto', 'min_samples_leaf': 2, 'min_samples_split': 5, 'splitter': 'random'}

Confusion Matrix

- Evaluated the **Decision Tree classifier** using a confusion matrix to assess its prediction performance.
- The model correctly classified the majority of **successful and unsuccessful** Falcon 9 landing outcomes.
- A small number of **false positives** occurred, where unsuccessful landings were incorrectly predicted as successful.
- **Key Insight:** Despite a few false positive predictions, the model demonstrated strong overall classification performance with **94.4% accuracy**, indicating good generalization on the test data.
- **Takeaway:** *The confusion matrix confirms that the Decision Tree model reliably distinguishes between successful and unsuccessful landings, making it suitable for predicting Falcon 9 first-stage landing outcomes.*



Conclusions

- Successfully developed an **end-to-end data science pipeline** to predict Falcon 9 first-stage landing success using data collection, data wrangling, exploratory analysis, visualization, and machine learning.
- **Launch site, orbit type, flight number, payload mass, and launch year** were identified as the most influential factors affecting landing success.
- The analysis showed that **landing success improved steadily between 2013 and 2020**, reflecting advancements in Falcon 9 technology and operational experience.
- **KSC LC-39A** achieved the highest landing success rate, while **ES-L1, GEO, HEO, SSO, and VLEO** missions demonstrated strong recovery performance.
- Among the evaluated models, the **Decision Tree classifier** achieved the **highest accuracy (94.4%)**, making it the most effective model for predicting first-stage landing success.

Future Work

- Incorporate additional features such as **weather conditions, booster age, and recent launch data** to further improve prediction accuracy.
- Evaluate advanced ensemble models such as **Random Forest** and **XGBoost** for enhanced predictive performance.

Thank you!

